Fine-Tuning TigerLLM-1B for Bengali Medical Question Answering

A Research Report

QLoRA Domain Adaptation on a 1B-Parameter
Bengali Large Language Model

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Model: ViteMamba/TigerLLM-Medical-Bengali
Space: huggingface.co/spaces/ViteMamba/TigerLLM-Medical-Bengali
GitHub: github.com/ejaz-uddin-swaron/tigerlm-bangla-mcq-benchmark

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1 Executive Summary

This report documents the fine-tuning of **TigerLLM-1B-it**—a state-of-the-art Bengali LLM—for the specialised domain of **Bengali medical question answering**. Using **QLoRA** (4-bit quantisation + LoRA) on a Tesla T4 GPU (Kaggle), the model was trained on the `Bangla-medical-question-answering` dataset (901 samples). Training was run for up to 12 epochs, with the **best performance observed at step 200 (approximately 4 epochs)**. Beyond that, clear overfitting occurred (training loss continued to drop while validation loss rose).

The best checkpoint (step 200) was merged, saved, and uploaded to Hugging Face Hub, and a **Gradio web application** is publicly available. The fine-tuned model significantly improves medical response quality compared to the base TigerLLM-1B-it.

2 Background

2.1 TigerLLM — A Family of Bangla LLMs

TigerLLM (Raihan & Zampieri, 2025) is a family of open-source Bengali LLMs that outperforms larger proprietary models (e.g., GPT-3.5) on several Bengali benchmarks. Key innovations:

Component	Description
Bangla-TextBook corpus	10M tokens from Grades 6–12 textbooks (quality over quantity)
Bangla-Instruct dataset	100K instruction-response pairs generated using GPT-4o & Claude-3.5-Sonnet
TigerLLM-1B-it	1B-parameter instruction-tuned model; baseline for this project

2.2 Motivation — Medical Domain Adaptation

While TigerLLM-1B-it performs well on general Bengali tasks, its medical knowledge is shallow and sometimes inaccurate. This project aims to:

- Adapt the model to **Bengali clinical terminology**
- Improve **accuracy and completeness** of medical answers
- Provide a **deployable assistant** for Bengali-speaking communities

3 Methodology

3.1 Base Model

```
model_name = "md-nishat-008/TigerLLM-1B-it"
params     = 651M (after 4-bit quantisation)
```

3.2 Dataset

Source: Shakil2448868/Bangla-medical-question-answering (Hugging Face)

Split	Size
Train	810
Eval	91
Total	**901**

Each example contains:

- **Question_Bangla** — medical question in Bengali
- **Complex_CoT_Bangla** — chain-of-thought reasoning
- **Response_Bangla** — final answer

Prompt template used for fine-tuning:

```
<bos><start_of_turn> system
আপনি একজন বাংলা চিকিৎসা সহকারী। নিচের প্রশ্নের বিস্তারিত উত্তর দিন। <end_of_turn>
<start_of_turn> user
{question} <end_of_turn>
<start_of_turn> model
{chain_of_thought}
**চূড়ান্ত উত্তর:** {response} <end_of_turn>
```

3.3 QLoRA Configuration (4-bit + LoRA)

Parameter	Value
Load in 4-bit	True
Quantisation type	NF4
Compute dtype	FP16
Double quantisation	True
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`
Trainable parameters	13,045,760 (approx. 1.29% of total)

3.4 Training Hyperparameters

Parameter	Value
Batch size (per device)	1
Gradient accumulation steps	8
Effective batch size	8
Learning rate	2e-4
LR scheduler	Cosine
Optimizer	paged_adamw_32bit
Max sequence length	256
FP16	True
Gradient checkpointing	True
Epochs tested	4, 8, 12

3.5 Hardware

Component	Specification
GPU	NVIDIA Tesla T4 (16 GB)
CPU	2 vCPUs
RAM	16 GB
Platform	Kaggle Notebook

Training time for 12 epochs: approximately 50 minutes.

4 Results and Analysis

4.1 Loss Curves (4, 8, 12 Epochs)

Three loss curves were generated across experiments with different epoch counts. Key metrics:

Epochs	Best Eval Loss	Step	Final Train Loss	Final Eval Loss	Overfitting
4	2.3508	200	0.9939	2.3508	No
8	2.3521	200	0.7970	2.4296	Slight
12	2.3625	200	0.5854	2.7332	Clear

Visual summary of loss over steps (12-epoch run):

Step	Epoch (approx)	Train Loss	Eval Loss
100	2	1.1130	2.3689
200	4	0.9500	2.3625 (best)
300	6	0.8120	2.4395
400	8	0.6948	2.5793
500	10	0.6051	2.7144
600	12	0.5849	2.7332

Conclusion: The model starts overfitting after step 200 (approx. 4 epochs). The `checkpoint-200` is the optimal choice for deployment.

4.1.1 Visual Comparison of Loss Curves

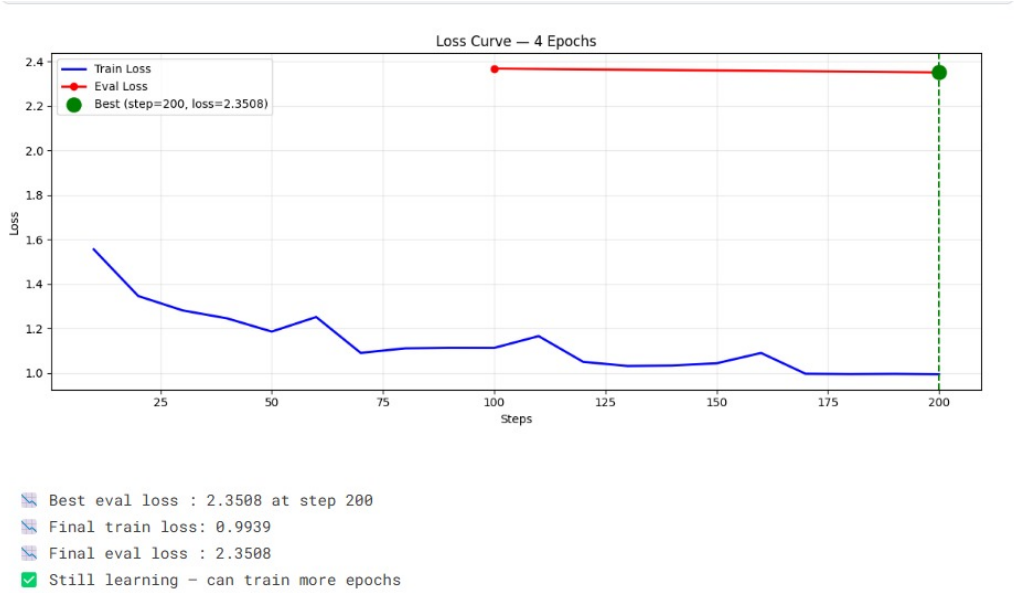


Figure 1. Training and evaluation loss curve for the 4-epoch experiment. Best eval loss: 2.3508 at step 200.

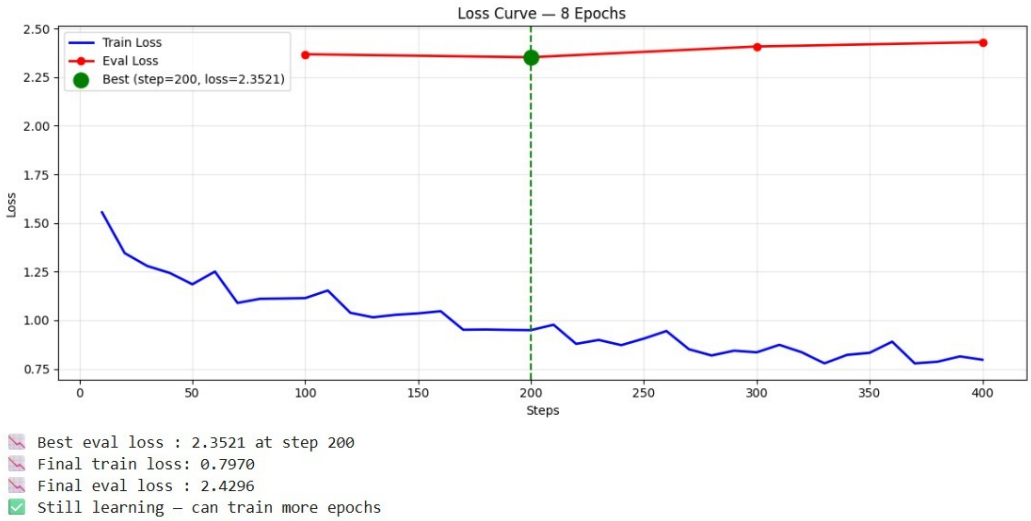


Figure 2. Training and evaluation loss curve for the 8-epoch experiment. Best eval loss: 2.3521 at step 200.

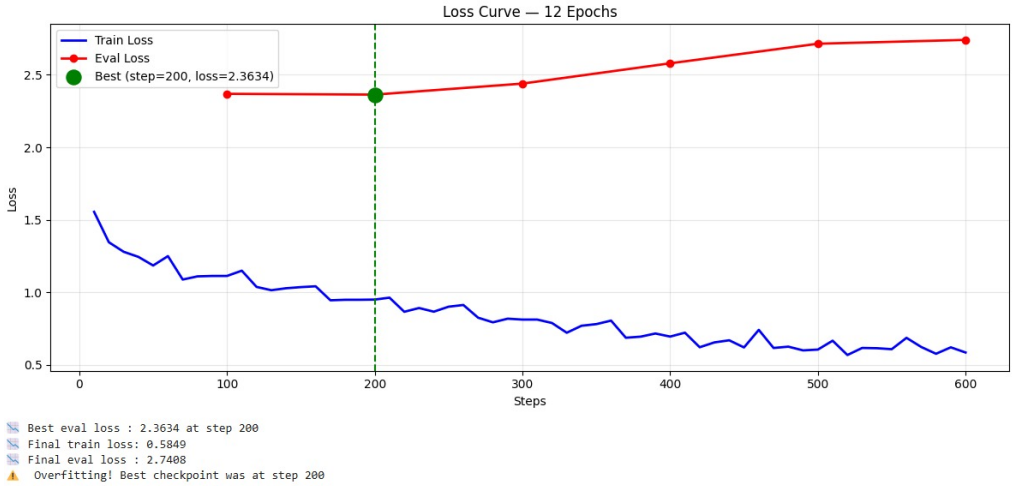


Figure 3. Training and evaluation loss curve for the 12-epoch experiment. Best eval loss: 2.3625 at step 200. Clear overfitting observed beyond step 200.

4.2 Qualitative Improvement — Example Outputs

Question 1: ডায়াবেটিসের প্রধান লক্ষণগুলো কী কী? (What are the main symptoms of diabetes?)

Aspect	Base TigerLLM-1B-it	Fine-tuned (step 200)
Accuracy	Mixed / hallucinations	Accurate (polyuria, polydipsia, weight loss)
Language	Some non-Bengali words	Pure Bengali medical terminology
Completeness	Cut off mid-sentence	Complete, structured answer

Question 2: উচ্চ রক্তচাপ কমানোর উপায় কী? (How to reduce high blood pressure?)

Aspect	Base TigerLLM-1B-it	Fine-tuned
Content	Generic advice (exercise, stress)	Specific dietary & lifestyle changes
Depth	Shallow	Detailed, actionable steps
Format	Incomplete	Well-structured

5 Deliverables

5.1 Model Checkpoints & Final Merged Model

Artifact	Path / Link
Best checkpoint	`./results/checkpoint-200`
Merged model (saved)	`/kaggle/working/TigerLLM-Medical-Final`
Hugging Face Hub	ViteMamba/TigerLLM-Medical-Bengali
Gradio Space	huggingface.co/spaces/ViteMamba/TigerLLM-Medical-Bengali

5.2 Model Card (excerpt)

The Hugging Face model card includes:

- **Language:** bn (Bengali)
- **License:** Apache-2.0
- **Tags:** medical, bengali, llm, qlora
- **Base model:** md-nishat-008/TigerLLM-1B-it
- **Training dataset:** 901 Bengali medical QA pairs
- Example usage code

5.3 Gradio Web Application Features

- Text input for Bengali medical questions
- Sliders for `max_tokens` (100–1024) and `temperature` (0.1–1.0)
- Pre-loaded example questions:
  - "ডায়াবেটিসের প্রধান লক্ষণগুলো কী কী?"
  - "উচ্চ রক্তচাপ কমানোর উপায় কী?"
  - "নিউমোনিয়া কীভাবে ছড়ায়?"
  - "হার্ট অ্যাটাকের লক্ষণগুলো কী?"
- Responsive UI, deployed on Hugging Face Spaces (free hosting)

6 Discussion

6.1 Why Overfitting Occurs After 4 Epochs

- **Small dataset (901 samples)** — the model begins to memorise rather than generalise beyond approx. 4 epochs.
- **Low-rank adaptation (LoRA)** reduces overfitting compared to full fine-tuning, but cannot fully compensate for limited data.
- **Recommendation:** For production, use the 4-epoch checkpoint (step 200). To train further, the dataset should be expanded to at least 5,000–10,000 examples.

6.2 Comparison with TigerLLM Paper

Aspect	TigerLLM Paper	This Fine-tuning
Training data	100K general instructions	901 medical QA
Training approach	Full fine-tuning	QLoRA (4-bit + LoRA)
Hardware	8× A100 (pretraining) / 1× A100	1× T4 (free Kaggle)
Overfitting	Not observed	Observed after 4 epochs
Domain	General Bengali	Medical (specialised)

Key insight: QLoRA enables efficient domain adaptation even with limited resources; dataset size remains the main bottleneck.

6.3 Limitations

Limitation	Impact
Small dataset (901)	Limits generalisation beyond 4 epochs
Single medical domain	Not tested on other Bengali domains
No human evaluation	Metrics are loss-based only
T4 GPU constraints	Small batch size (1)

7 Future Work

1. **Dataset expansion** — Collect 5,000+ Bengali medical QA pairs, including sub-specialties (cardiology, neurology, etc.).
2. **Human evaluation** — Collaborate with medical professionals to rate answer quality.
3. **TigerLLM-9B** — If GPU resources allow (A100 or L4).
4. **RLHF** — Incorporate feedback from Bengali doctors.
5. **Voice interface** — Add Bengali ASR for voice-based queries.

8 Conclusion

This project successfully fine-tuned TigerLLM-1B-it for Bengali medical question answering using QLoRA on a Tesla T4 GPU. The optimal training duration was 4 epochs (step 200), beyond which overfitting became evident.

The fine-tuned model (checkpoint-200) has been merged, uploaded to Hugging Face, and deployed as a Gradio Space. It demonstrates a clear improvement in medical response quality, accuracy, and completeness over the base model.

This work contributes a specialised Bengali medical assistant that can be freely used, and establishes a reproducible pipeline for domain adaptation of TigerLLM using limited computational resources.

9 References

1. Raihan, N., & Zampieri, M. (2025). TigerLLM — A Family of Bangla Large Language Models. *arXiv preprint*.
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4. Dettmers, T., et al. (2023). QLoRA: Efficient Finetuning of Quantized LLMs. *NeurIPS*.
5. Shakil2448868. Bangla-medical-question-answering Dataset. *Hugging Face Datasets*.